PROBLEM SET 11 DUE 6.13

1. Let $A \subseteq B$ be rings and let C the integral closure of A in B. Let S be a multiplicative subset of A. Show that $S^{-1}C$ is the integral closure of $S^{-1}A$ in $S^{-1}B$.

2. Let A be an integral domain. Show that the following are equivalent:

(1) A is integrally closed.
(2) $A_{\mathfrak{p}}$ is integrally closed for every $\mathfrak{p} \in \operatorname{Spec} A$.
(3) $A_{\mathfrak{m}}$ is integrally closed for every $\mathfrak{m} \in \operatorname{Specm} A$.

Hint: Use Problem **1** *and Problem Set* **10** *Problem* **1**.

3. Let k be a field, $\operatorname{char} k \neq 2$, and t an indeterminate. Consider the ring extension

$$A \coloneqq k[t^2] \subseteq k[t] =: B$$

Obviously, B is integral over A. For $a \in k$, denote the maximal ideal

$$\mathfrak{n} \coloneqq (t-a) \subseteq B$$

and put

$$\mathfrak{m} = \mathfrak{n} \cap A.$$

Prove the following

(1) If $a \neq 0$, then $B_{\mathfrak{n}}$ is **not** integral over $A_{\mathfrak{m}}$.

Hint: Show that $\frac{1}{t+a}$ *is not integral over* $A_{\mathfrak{m}}$.

(2) If $a = 0$, then $B_{\mathfrak{n}}$ is integral over $A_{\mathfrak{m}}$.

4. Let $A \subseteq B$ be an integral extensions of rings. Show that

$$A \cap \operatorname{rad}(B) = \operatorname{rad}(A)$$

where $\operatorname{rad}(A), \operatorname{rad}(B)$ are the Jacobson radicals of A, B respectively.

5. Let $A \subseteq B$ be an integral extension of rings and let $f : A \to \Omega$ be a homomorphism of A to an algebraically closed field Ω. Show that f can be extended to a homomorphism of B to Ω.

Hint: Use lying-over to reduce the problem to the case where both A and B are integral domains and f is injective. Show that if $A \subseteq B$ is an integral extension of integral domains, and $K = \operatorname{Frac}(A)$, $L = \operatorname{Frac}(B)$, then $K \subseteq L$ is an algebraic field extension.